

$[a, b] \rightarrow$ cerrado

$x_0 \rightarrow$ Abierto

Formulario Analisis Numérico

Error Absoluto: $E_A = |\text{Valor real} - \text{Valor truncado redondeado}|$

Error Relativo: $er = \frac{|E_A|}{\text{valor real}} * 100\%$

Punto Fijo 0.01%

Aproximaciones sucesivas

$$f(x) = \underbrace{\quad}_x = 0 \quad x = g(x) \quad |g'(x)| < 1$$

It x_i : $|f(x_i)| = |x_{i+1} - x_i| < 1$ er

Factores cuadráticos

$$P(x) = a_0 x^n + a_1 x^{n-1} + \dots + a_n \approx (x^2 + p x + q)_2$$

$n \Rightarrow$ grado polinomio $b_x^{n-2} \dots + b_{n-3} + b_{n-2} \rightarrow +R x + S$

It 1 $\rightarrow p_1 = q_1 = b_{n-1} = b_{n-2} = 0$

$$\Delta p_i = \frac{R}{b_{n-2}}$$

It 2 en adelante: $p_{i+1} = p_i + \Delta p_i$

$$R = a_{n-1} - p_i b_{n-2} \quad q_{i+1} = q_i + \Delta q_i \quad \Delta q_i = \frac{S}{b_{n-2}}$$

$$S = a_n - q_i b_{n-2}$$

$$b_k = a_k - p_i b_{k-1} - q_i b_{k-2}$$

$k = 0, \dots, n-2$

Bisección $[a, b]$ $C = \frac{a+b}{2}$

I $|a| f(a) |b| f(b) |c| f(c) |er$

Regla falsa

$$C = a + \frac{(b-a) f(a)}{f(b) - f(a)}$$

Newton-Raphson

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)} \quad g'(x_{i+1}) = \frac{f(x_{i+1}) \cdot f''(x_{i+1})}{(f'(x_{i+1}))^2}$$

It x_i : $|f(x_i)| < 1$ er

$$\text{Maclaurin: } P(x) = \sum_{k=0}^n \frac{(x-a)^k}{k!} f^{(k)}(a)$$